

SHETH LUJ AND SIR MV COLLEGE
Subject: Data Analysis with SAS / SPSS / R

M2 Practical No: 6

Aim: Performing paired t-tests using `t.test(paired=TRUE)` (R).

Code:

1. Load Data

```
rm(list = ls())
```

```
data <- read.csv("AmesHousing.csv")
```

```
data_paired <- data[c("X1st.Flr.SF", "X2nd.Flr.SF")]
```

```
names(data_paired) <- c("Flr_1_SF", "Flr_2_SF")
```

```
data_paired <- na.omit(data_paired)
```

2. Check structure

```
print("--- 2. Data Structure ---")
```

```
str(data_paired)
```

3. Descriptive statistics

```
print("\n--- 3. Descriptive Statistics ---")
```

```
mean(data_paired$Flr_1_SF)
```

```
mean(data_paired$Flr_2_SF)
```

```
sd(data_paired$Flr_1_SF)
```

```
sd(data_paired$Flr_2_SF)
```

4. Normality Test on Differences

```
difference <- data_paired$Flr_2_SF - data_paired$Flr_1_SF
```

```
print("\n--- 4. Normality Test on Differences (Shapiro-Wilk) ---")
```

```
shapiro.test(difference[1:500])
```

5. Paired t-test

```
print("\n--- 5. Paired t-test (Default Method) ---")
```

```
paired_t_test <- t.test(
```

```
  data_paired$Flr_1_SF,
```

```
  data_paired$Flr_2_SF,
```

```
  paired = TRUE
```

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)

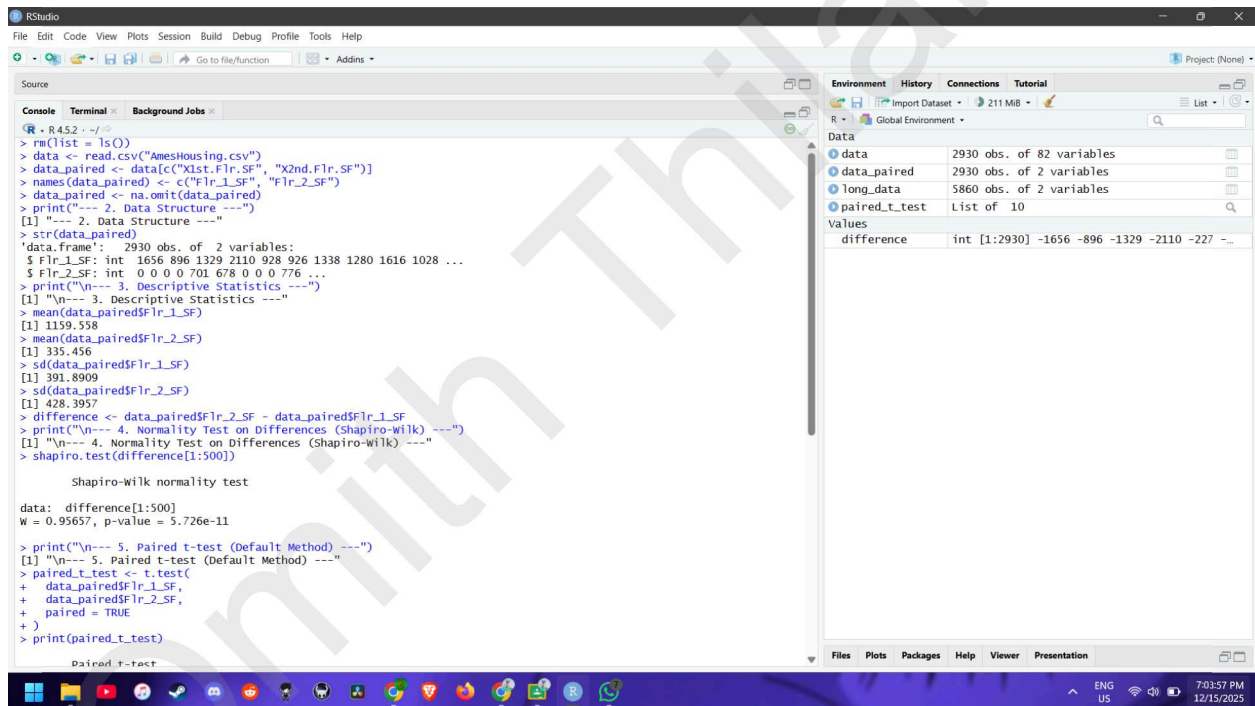
```
print(paired_t_test)
```

6. Alternative way (Formula method - using long format conversion)

```
long_data <- data.frame(  
  Score = c(data_paired$Flr_1_SF, data_paired$Flr_2_SF),  
  Time = rep(c("Flr_1_SF", "Flr_2_SF"), each = nrow(data_paired))  
)
```

```
print("\n--- 6. Alternative Paired t-test (Formula Method) ---")  
t.test(data_paired$Flr_1_SF, data_paired$Flr_2_SF, paired = TRUE)
```

Output:



The screenshot shows the RStudio interface with the following content:

```
R - R4.5.2 - ~/R  
File Edit Code View Plots Session Build Debug Profile Tools Help  
Source  
rm(list = ls())  
> data <- read.csv("AmesHousing.csv")  
> data_paired <- data[c("X1st.Flr_SF", "X2nd.Flr_SF")]  
> names(data_paired) <- c("Flr_1_SF", "Flr_2_SF")  
> data_paired <- na.omit(data_paired)  
> print("--- 2. Data Structure ---")  
[1] "--- 2. Data Structure ---"  
> str(data_paired)  
'data.frame': 2930 obs. of 2 variables:  
 $ Flr_1_SF: int 1656 896 1329 2110 928 926 1338 1280 1616 1028 ...  
 $ Flr_2_SF: int 0 0 0 0 701 678 0 0 0 776 ...  
> print("\n--- 3. Descriptive Statistics ---")  
[1] "\n--- 3. Descriptive Statistics ---"  
> mean(data_paired$Flr_1_SF)  
[1] 1159.558  
> mean(data_paired$Flr_2_SF)  
[1] 335.456  
> sd(data_paired$Flr_1_SF)  
[1] 391.8909  
> sd(data_paired$Flr_2_SF)  
[1] 428.3957  
> difference <- data_paired$Flr_2_SF - data_paired$Flr_1_SF  
> print("\n--- 4. Normality Test on Differences (Shapiro-wilk) ---")  
[1] "\n--- 4. Normality Test on Differences (Shapiro-wilk) ---"  
> shapiro.test(difference[1:500])  
  
Shapiro-wilk normality test  
data: difference[1:500]  
W = 0.95657, p-value = 5.726e-11  
  
> print("\n--- 5. Paired t-test (Default Method) ---")  
[1] "\n--- 5. Paired t-test (Default Method) ---"  
> paired_t_test <- t.test(  
+ data_paired$Flr_1_SF,  
+ data_paired$Flr_2_SF,  
+ paired = TRUE  
+ )  
> print(paired_t_test)
```

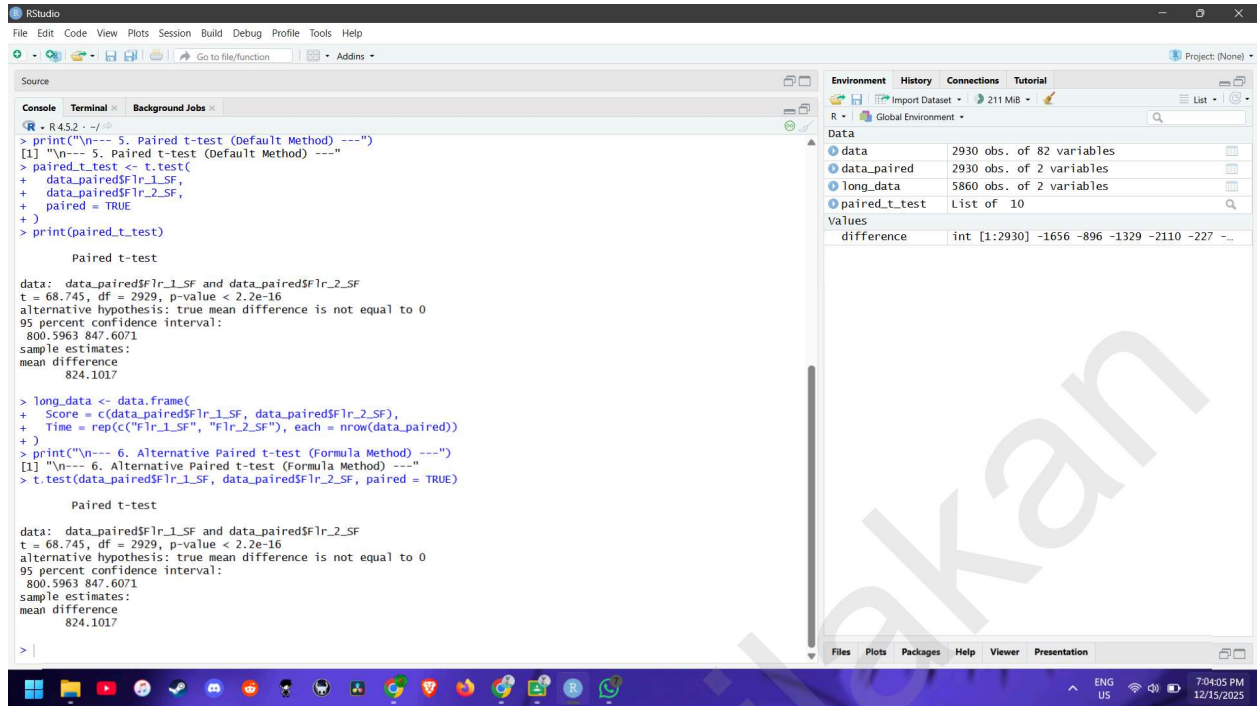
The Environment pane on the right shows the following objects:

Object	Details
data	2930 obs. of 2 variables
data_paired	2930 obs. of 2 variables
long_data	5860 obs. of 2 variables
paired_t_test	List of 10

The Values pane shows the following data:

difference	int [1:2930]
	-1656 -896 -1329 -2110 -227 ...

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```
> print("\n--- 5. Paired t-test (Default Method) ---")
[1] "\n--- 5. Paired t-test (Default Method) ---"
> paired_t_test <- t.test(
+   data_paired$flr_1_SF,
+   data_paired$flr_2_SF,
+   paired = TRUE
+ )
> print(paired_t_test)

Paired t-test

data: data_paired$flr_1_SF and data_paired$flr_2_SF
t = 68.745, df = 2929, p-value < 2.2e-16
alternative hypothesis: true mean difference is not equal to 0
95 percent confidence interval:
 800.5963 847.6071
sample estimates:
mean difference
 824.1017

> long_data <- data.frame(
+   Score = c(data_paired$flr_1_SF, data_paired$flr_2_SF),
+   Time = rep(c("flr_1_SF", "flr_2_SF"), each = nrow(data_paired))
+ )
> print("\n--- 6. Alternative Paired t-test (Formula Method) ---")
[1] "\n--- 6. Alternative Paired t-test (Formula Method) ---"
> t.test(data_paired$flr_1_SF, data_paired$flr_2_SF, paired = TRUE)

Paired t-test

data: data_paired$flr_1_SF and data_paired$flr_2_SF
t = 68.745, df = 2929, p-value < 2.2e-16
alternative hypothesis: true mean difference is not equal to 0
95 percent confidence interval:
 800.5963 847.6071
sample estimates:
mean difference
 824.1017

> |
```

Environment: R, Global Environment, 211 MiB

Data:

Object	Size
data	2930 obs. of 82 variables
data_paired	2930 obs. of 2 variables
long_data	5860 obs. of 2 variables
paired_t_test	List of 10

Values:

difference	int [1:2930]
	-1656 -896 -1329 -2110 -227 ...